

Dr. Soumyabrata Chakravarty

Ph.D (Engineering), Jadavpur University

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Professional Summery

Research-driven and policy-aware professional with 5+ years of experience in academia and clean technology R&D. Demonstrated record in sustainable manufacturing, renewable energy materials, and circular economy. Adept at stakeholder engagement, technical writing, datadriven research, and clean energy systems analysis. Committed to contributing to India's inclusive green energy transition through evidence-based policy work and multi-stakeholder collaboration.

Professional Experience

Assistant Professor

Brainware University, Kolkata | Aug 2024 – Present | 2 Years 1 Month

- Teaching undergraduate and diploma-level engineering courses in Energy, Manufacturing, and Design.
- Active contributor to institutional processes like NAAC, academic audits, and curriculum development.
- Mentoring student projects in sustainability and clean energy innovations.

Teaching Assistant (Contractual)

NITTTR, Kolkata | Dec 2023 – Jul 2024 | 7 Months

- Designed academic modules focusing on Indian Knowledge Systems in metallurgy and ancient manufacturing.
- Conducted desktop research and literature reviews on metalworking technologies and sustainable practices.

Senior Research Fellow – Dept. of Science, Technology & Biotechnology, Govt. of West Bengal (WBDST&BT)

Mechanical Engineering Dept., Jadavpur University | Nov 2022 – Nov 2023 | 12 Months

- Led experimental work on aluminium metal matrix composites using cupola slag, a sustainable reuse strategy.
- Published impactful papers on material circularity and green manufacturing techniques.
- Engaged with stakeholders on sustainable waste management and circular economy themes.

Junior Research Fellow – Dept. of Science, Technology & Biotechnology, Govt. of West Bengal (WBDST&BT)

Mechanical Engineering Dept., Jadavpur University | Nov 2022 – Nov 2023 | 24 Months

- Conducted experimental and analytical studies on reuse of industrial waste in composite manufacturing.
- Gained experience in technical research and state-level environmental project execution.

Project Assistant – Rashtriya Uchattar Shiksha Abhiyan (RUSA 2.0)

Power Engineering Dept., Jadavpur University | Jul 2019 – Mar 2020 | 10 Months

- Worked on laser cladding optimization under the Central Govt. RUSA 2.0 scheme.
- Conducted parameter analysis for energy-efficient surface engineering.

Key Skills

- **Clean Energy & Circular Economy:** Waste-to-resource research, distributed energy systems, green manufacturing
- **Research Tools:** Excel (Advanced), Minitab, COMSOL
- **Technical Proficiency:** CATIA V6, SolidWorks, AutoCAD, MS Office, C++
- **Stakeholder Collaboration:** Experience presenting in national forums, working with academia, industry, and government bodies

Publications & Research Impact

- 15 Journal Articles (10 SCI, 3 ESCI, 2 SCOPUS), 2 Book Chapters
- Over 130 citations, h-index: 7 | [Google Scholar Profile](#)
- Key Themes: Renewable material composites, energy optimization, sustainable industrial practices, Manufacturing, Material science

Selected Awards & Presentations

- **Soli Commissariat Award** – Best Paper, 73rd Indian Foundry Congress, 2025
- Presented research at conferences including **COPEN-IIT Indore**, **ICSTE-NIT Manipur**, and **INCOM-JU**
- Contributed to national dialogue on industrial waste reuse and green composite development

Education

Ph.D. in Mechanical Engineering

Jadavpur University, 2025

Thesis Title: Machinability study on cupola slag reinforced aluminium metal matrix composites

- *Developed sustainable AMCs using cupola slag waste, achieving 70% higher tensile strength and enhanced machinability while reducing material cost and weight.*
- *Innovated lightweight aluminium composites via industrial waste reinforcement, delivering 47% higher hardness and significant grain refinement using stir casting.*

M.Tech in Laser Science and Technology – 9.17 CGPA

Jadavpur University, 2019

B.Tech in Mechanical Engineering – 8.24 CGPA

WBUT, 2016

Workshops & Training

- TEQIP Workshop on Optimization in Advanced Manufacturing (VSSUT, Odisha)
- Technical Training at SAIL-IISCO, Burnpur & ATI, Howrah
- Certified in CATIA V6, SolidWorks (MSME Tool Room)

References

- **Prof. Titas Nandi**
Professor, Mechanical Engineering, Jadavpur University *titas_nandi@yahoo.in*
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 - **Prof. Goutam Sutradhar** Director, NIT Jamshedpur
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Annexure: List of Publication

Journals

1. **S. Chakravarty**, S. Mondal, & D.P. Mishra, A critical review on wootz steel of India. *Indian Journal of History of Science*, 61(1), 88-107 (2026) [**Q3 ESCI, IF: 0.9**]
2. **S. Chakravarty**, R. Sikder, P. Haldar, T. Nandi and G. Sutradhar. Dry turning behavior of solution heat treated cupola slag reinforced aluminum matrix composites. *Physica Scripta*, 100(5), 055925 (2025) [**Q2, SCI, Impact Factor: 2.6**]
3. **S. Chakravarty**, P. Haldar, T. Nandi and G. Sutradhar. Enhancement of mechanical properties and machinability of aluminium composites by cupola slag reinforcements. *Materialwissenschaft und Werkstofftechnik (Materials Science and Engineering Technology)*. 55, 1250 (2024) [**Q3, SCI, Impact Factor: 1.1**]
4. **S. Chakravarty**, R. Sikder, P. Haldar, T. Nandi and G. Sutradhar. Exploring energy aspects and tool wear on dry turning of cupola slag-reinforced aluminium metal matrix composites. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*. 46, 59 (2024). [**Q2, SCI, Impact Factor: 2.2**]
5. **S. Chakravarty**, R. Sikder, P. Haldar, T. Nandi and G. Sutradhar. Machinability of Cupola Slag Incorporated LM11 Matrix Composites in Turning: Surface Roughness and MRR Analysis. *Materials Today Communications*, 107414, (2023) [**Q2, SCI, Impact Factor: 3.8**]
6. **S. Chakravarty**, R. Sikder, P. Haldar, T. Nandi and G. Sutradhar. Experimental investigation on feasibility of industrial waste to resource conversion for cupola slag. *Results in Engineering*, 17, 100962, (2023) [**Q1, ESCI, Impact Factor: 5**]
7. **S. Chakravarty**, P. Haldar, T. Nandi and G. Sutradhar. Cupola slag reutilization for sustainable waste management: review and economic analysis. *International Journal of Environmental Science and Technology*, 20, 1169–1184 (2021). [**Q1, SCI, Impact Factor: 3.1**]
8. R. Sikder, **S. Chakravarty**, D. Sao, P. Haldar, S. Mandal, T. Nandi and G. Sutradhar. Study of durability and mechanical properties for utilization of cupola slag as a coarse aggregate in M20 grade green concrete, *Journal of Building Engineering*, Volume 87, 109101 (2024) [**Q1, SCI, Impact Factor: 6.4**]
9. R. Sikder, **S. Chakravarty**, P. Haldar, T. Nandi and G. Sutradhar. Scope for cupola slag reuse in construction: a sustainable green solution. *International Journal of Environmental Science and Technology*, 20(6), 7019-7028, (2023). [**Q1, SCI, Impact Factor: 3.1**]
10. R. Sikder, D. Sau, **S. Chakravarty**, P. Haldar, S. Mandal, T. Nandi, & G. Sutradhar, Using cupola furnace slag in concrete: A sustainable approach to protect natural aggregates and the environment. *Environmental Progress & Sustainable Energy*, e70174. (2025) [**Q2, SCI, Impact Factor: 2.3**]
11. S. Hazra, S. Chattopadhyay, **S. Chakravarty**, and N. K. Mandal, Tribology of Mg-B₄C nanocomposites fabricated following powder metallurgy route. *Engineering Research Express*, 6, 035525, (2024). [**Q2, ESCI, Impact Factor: 1.6**]

- 12.N. Kumar, **S. Chakravarty**, N. Kumar and R. S. Sen, A Combined RSM – FEM Analysis of Weld Quality in Laser Transmission Welding of Plastics. *Materials Research Express*, 11(11), 115310, (2024). [**Q2, SCI, Impact Factor: 1.8**]
- 13.H. Kumar, **S. Chakravarty**, N. Kuamr, & N. Kumar, Prediction of laser welding qualities of Al alloys using regression and machine learning techniques. *Materials Research Express*, 12(6), 066501, (2025). [**Q2, SCI, Impact Factor: 1.8**]
- 14.A. K. Panja, B. Bose, **S. Chakravarty**, A. Das Sarma, D. Bhowmick, & N. Kumar, Fabrication and Mechanical Characterization of Hybrid Polymer Matrix Composites. *Engineering Research Express*. 7 035542, (2025) [**Q2, ESCI, Impact Factor: 1.6**]
- 15.S. Sarkar, **S. Chakravarty**, B. Barai, I. Koley, S. Ghuku, P. Biswas, &P. Haldar, Optimization of Corrosion Rate of Coatings Using RSM and Particle Swarm Optimization Algorithm. *Journal of Molecular and Engineering Materials*, 10(03n04), 2250004, (2022). [**SCOPUS**]
- 16.M. N.Ali, **S. Chakravarty** and P. Haldar Experimental investigation and optimization of MRR in μ -ECDM process by taguchi, RSM, PSO and ANN. *Suranaree Journal of Science & Technology*, 29(5) (2022). [**Q4, SCOPUS**]

Conference Publications

1. **S. Chakravarty**, P. Haldar, T. Nandi and G. Sutradhar. Fabrication and machinability studies on cupola slag reinforced aluminium metal matrix composites using Taguchi method. *Materials Today: Proceedings*, (2023). [**SCOPUS**]
2. R. Sikder, **S. Chakravarty**, P. Haldar, T. Nandi and G. Sutradhar. Cupola slag as partial replacement of coarse aggregate in concrete. *Materials Today: Proceedings*, (2023). [**SCOPUS**]

Book Chapters

1. **S. Chakravarty**, P. Haldar, T. Nandi and G. Sutradhar.Fuzzy Logic-Based Model for Predicting Material Removal Rate of Machined Cupola Slag-Reinforced Aluminum Metal Matrix Composite. *Recent Advances in Materials. ICSTE 2023. Springer Proceedings in Materials*, vol 25. Springer, (2023).
2. **S. Chakravarty**, P. Haldar and N. Modak. Testing and Characterization of Fiber Reinforced Epoxy and Polymer Composite, Elsevier, (2022).